

Animal Behavior Observation Lab

Answer Key / Grading Guide (10 points total)

TA note: As with the original “wild primate” exercise, **answers will vary**. Grade for correctness of scientific reasoning, appropriate use of the chi-square test, and internal consistency — not for matching specific values.

This grading scale is intentionally **isomorphic** to *Exercise 2* in the Lab 1.2 answer key you provided.

Introduction / Hypotheses (4 points total)

1. Hypotheses (3 pts total)

Students should write hypotheses for **one chi-square test of independence** comparing two conditions.

Null hypothesis (H) — 1.5 pts

- States **no difference** in behavior distribution between Condition A and Condition B
- Mentions **behavior categories** (dependent variable)
- Mentions **conditions** (independent variable)

Example (wording may vary):

“The proportion of observed behavior categories is the same between the first and second 10 minutes of observation.”

Alternative hypothesis (H) — 1.5 pts

- States that **behavior distribution differs** between conditions
- Correctly framed as the opposite of H

Example:

“The distribution of observed behaviors differs between Condition A and Condition B.”

2. Research question (1 pt)

- Clearly stated
- Logically matches the hypotheses and test performed

Examples:

- “Does animal behavior differ between the first and second half of the observation period?”
- “Does behavior differ between two animals in the same exhibit?”

Methods (3 points total)

This section replaces the multiple-location structure of the original lab while preserving its logic.

3. Variables (1.5 pts)

- **Dependent variable (0.5 pts):** Animal behavior category (from ethogram)
- **Independent variable (1.0 pt):** Comparison condition (time period, animal identity, or exhibit zone)

TA note: This maps directly onto the original lab’s “*Observed vs expected*” and “*different locations*” framing.

4. Sample size and replication (1.5 pts)

- **Sample size stated correctly (0.75 pts):** $n = 10$ scans per condition (20 total)
- **Replication discussion (0.75 pts):** Acceptable answers:
 - No replication within individual
 - Replication occurs across students/groups observing similar systems

This matches the original lab's language exactly.

Results and Discussion (3 points total)

5. Chi-square results and interpretation (3 pts)

Students must:

- Correctly perform a **chi-square test of independence**
- Interpret the result biologically (not just statistically)

Breakdown:

a. Correct test structure (1 pt)

- Uses observed vs expected counts
- Uses correct chi-square formula
- Uses correct $df = 4$

b. Statistical decision (1 pt)

- Correctly states:
 - χ^2 value
 - p-value (table or software)
 - Reject or fail to reject H_0
- Decision must be **consistent** with results

TA note: Significance is *not required* — consistency is.

c. Biological interpretation (1 pt)

- Describes **which behaviors were common vs rare**
- Avoids anthropomorphism
- Explains what the result suggests about behavior in the observed context

Example:

“Inactive behavior was observed more frequently than expected in the second half of the observation, suggesting increased resting over time.”

Observations for Future Experiments (2 points total)

6. Improvements / limitations (2 pts)

Students must list **at least two** reasonable ideas.

Each valid point = **1 pt**

Acceptable answers include:

- Longer observation period
- Multiple days or times
- More individuals
- Fewer or merged behavior categories
- Improved camera angles
- Using focal sampling instead of scan sampling

This directly mirrors the original lab.

Point Summary (Exact Match to Existing Lab)

Section	Points
Hypotheses (H + H)	3
Research question	1
Variables	1.5
Sample size & replication	1.5
Results & interpretation	3
Future improvements	2
Total	10